

ADAPTIVE DIGITAL SAT PRACTICE TEST

SAT Practice Test 4 - Standardized Edition

Section 1: Reading and Writing

Module 1 (27 questions) | Module 2 Easier (27) | Module 2 Harder (27)

Section 2: Math

Module 1 (22 questions) | Module 2 Easier (22) | Module 2 Harder (22)

Total: 147 questions

Section 1: Reading and Writing

Module 1 (27 Questions) - All Students

Time: 32 minutes

Question 1 Skill: Words in Context | Difficulty: Medium

The following text is from Langston Hughes's 1926 poem "The Negro Speaks of Rivers." The speaker reflects on the deep history of African heritage: "I've known rivers ancient as the world and older than the flow of human blood in human veins. My soul has grown deep like the rivers." The speaker's tone throughout is one of quiet gravity, connecting personal identity to a vast historical lineage.

As used in the text, what does the word "gravity" most nearly mean?

- A. Heaviness
- B. Seriousness
- C. Attraction
- D. Descent

Question 2 Skill: Words in Context | Difficulty: Medium

The following text is from a 1947 translation of Albert Camus's 1942 novel *The Stranger* (translated by Stuart Gilbert). In the text, Meursault is observing the sea from a hillside above the beach. The glare of the afternoon sun was almost unbearable, and the light

seemed to press against his eyes with a physical force, as though the sky itself were leaning down upon the coastline, heavy and immovable.

As used in the text, what does the word “press” most nearly mean?

- A. Publish
- B. Urge
- C. Push
- D. Iron

Question 3 Skill: Words in Context | Difficulty: Medium

When astronomers detect a new comet, it is assigned a provisional designation based on the year and half-month of discovery, along with a sequential letter. After the comet’s orbit has been confirmed through multiple observations, the discoverers may propose a permanent name. Under current International Astronomical Union guidelines, it is _____ for discoverers to name comets after themselves, but they are not obligated to do so; the final naming decision rests with the relevant committee.

Which choice completes the text with the most logical and precise word or phrase?

- A. permissible
- B. infeasible
- C. selective
- D. mandatory

Question 4 Skill: Words in Context | Difficulty: Medium

The intricate ceramic mosaics adorning the Alhambra Palace in Granada, Spain, demonstrate the Moorish contribution to Iberian architecture, an influence that is _____ throughout the region—visitors touring the Mezquita in Córdoba, for instance, can readily identify how its design draws inspiration from North African building traditions.

Which choice completes the text with the most logical and precise word or phrase?

- A. corroborated
- B. palpable
- C. disputed
- D. understated

Question 5 Skill: Overall Structure | Difficulty: Medium

The following text is from Anton Chekhov’s 1899 short story “The Lady with the Dog” (translated by Constance Garnett in 1917). The narrator is describing a gathering at a seaside resort. A gentleman accompanied by his wife arrived at that moment and seated himself at the table nearest the orchestra. He surveyed the dining room with the air of a proprietor, as if to say, “I have dined in finer establishments across Europe, and you are all fortunate that I have chosen to grace this provincial hall with my presence.” There are travelers who approach foreign resorts as conquerors rather than guests, and it is well known that such persons rarely find satisfaction in any destination.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A. It conveys the eagerness some resort guests feel to be noticed by the other diners.
- B. It emphasizes a character's desire to become a restaurant owner.
- C. It helps explain why some travelers avoid seaside resorts altogether.
- D. It illustrates the gentleman's condescending attitude toward the other people in the dining room.

Question 6 Skill: Overall Structure | Difficulty: Medium

Dune, first published in 1965, is a novel in Frank Herbert's science fiction series, which includes six novels by Herbert plus numerous sequels by other authors. Like the rest of the books in the series, *Dune* has a rather complex ending, but the following book, *Dune Messiah*, picks up directly where *Dune* leaves off. Thus, the complexity of the ending is only an issue if one considers the books as independent texts—the series is best thought of as a single sprawling narrative, similar to other multivolume epics, such as Isaac Asimov's *Foundation* series.

Which choice best describes the function of the underlined portion in the text as a whole?

- A. It demonstrates that those readers who most enjoy reading *Dune Messiah* will be those who have just finished reading *Dune*.
- B. It argues that Herbert's series deserves the critical acclaim of works like the *Foundation* series.
- C. It offers a detail that helps explain why it is not necessarily the case that a certain characteristic of *Dune* should be considered a flaw.
- D. It presents a reason *Dune* and *Dune Messiah* are considered to be two of Herbert's most engaging novels.

Question 7 Skill: Overall Structure | Difficulty: Medium

Japanese architect Tadao Ando's prolific career, which spanned the 1970s to the 2020s, evolved through distinct phases. As epitomized by the Row House in Sumiyoshi, many of Ando's early projects integrated minimalist concrete forms into densely packed urban neighborhoods. Extensive travels abroad later sparked an engagement with sacred and contemplative spaces—designs whose emphasis on light, water, and silence and whose departure from purely residential commissions are readily apparent in Ando's Church of the Light in Ibaraki.

Which choice best describes the function of the underlined portion in the text as a whole?

- A. It presents biographical information that accounts for the transition in Ando's career that is discussed in the text.
- B. It notes why the two architectural projects described in the text constitute major departures from Ando's typical style.
- C. It lends support to an argument about the minimalist design aesthetic that is made in the previous sentence.
- D. It explains why the architectural project discussed later in the sentence was highly regarded by Ando's contemporaries.

Question 8 Skill: Main Idea | Difficulty: Medium

Colombian muralist Sandra Morales transforms public spaces using her distinctive method of layering botanical illustrations over photographic portraits of community members. In some works, Morales focuses on honoring historical figures who shaped civil rights, as she does in her mural of Berta Cáceres. However, in other works,

Morales celebrates unrecognized individuals, as she does in her striking depiction of an elderly fisherman mending nets by a river. Morales views both of these approaches as ways of representing the resilience and interconnectedness of all communities.

Which choice best states the main idea of the text?

- A. Morales began her artistic career by painting murals of historical figures and then transitioned to depicting ordinary community members instead.
- B. Morales's portrayal of an elderly fisherman focuses more on communal bonds than it does on individual hardship.
- C. Morales's works focus on honoring both prominent and everyday examples of human resilience and connection.
- D. Morales's depiction of Berta Cáceres receives more scholarly attention than her depiction of the elderly fisherman does.

Question 9 Skill: Main Idea | Difficulty: Medium

Why do certain species of frogs produce loud calls while others rely on visual displays? Researchers hypothesize that this difference may be partly due to the structure of the vocal sac, a specialized organ in the throat. Tree frogs, which are generally small, have a highly elastic vocal sac that vibrates rapidly when inflated, producing a sharp, resonant call. By contrast, poison dart frogs have a relatively stiff vocal sac and instead rely on bright coloration and rhythmic limb movements. These traits allow tree frogs and many other species of arboreal frogs to produce powerful calls. The same traits may also make it difficult for these frogs to use visual signaling.

Which choice best states the main idea of the text?

- A. Both tree frogs and poison dart frogs have vocal sacs, but tree frogs' sacs are elastic whereas poison dart frogs' sacs are stiff.
- B. Tree frogs and most other arboreal frog species can produce loud calls because they have elastic vocal sacs.
- C. Researchers are uncertain why relatively small frogs, such as tree frogs, call loudly but relatively colorful frogs, such as poison dart frogs, use visual displays.
- D. Differences related to the vocal sac in tree frogs and poison dart frogs may help explain why one species calls and the other uses visual displays.

Question 10 Skill: Main Idea | Difficulty: Medium

In their systematic review of research on branded mobile applications (apps developed by companies to promote products or services), Priya Mehta et al. confirm that such apps, though they can boost user engagement, may not translate into increased purchase intent or brand loyalty. This phenomenon can be explained by the finite nature of attentional resources as it is described in Daniel Kahneman's dual-process theory of cognitive load. In this case, users' cognitive resources are directed primarily toward the app's features, leaving little or no capacity for processing and retaining the marketing messages the developer intends to convey.

Which choice best states the main idea of the text?

- A. Kahneman's dual-process theory provides a framework for explaining the finding by Mehta et al. that users may not readily absorb the marketing messages embedded in branded apps.
- B. The systematic review by Mehta et al. reveals that higher engagement in branded app features is linked to lower

- effectiveness in persuading users to purchase particular products.
- C. Although Kahneman’s dual-process theory suggests otherwise, branded apps can succeed as marketing tools, provided that they balance app functionality and brand promotion.
 - D. Research by Mehta et al. supports Kahneman’s conclusion that because people primarily focus on app functionality, it is difficult for branded apps to communicate marketing messages memorably.

Question 11 Skill: Rhetorical Synthesis | Difficulty: Medium

Portable Music Players of the 1990s and 2000s

Device	Manufacturer	Device Type	Approx. Units Sold Worldwide
MiniDisc Walkman	Sony	disc player	2,200,000
Diamond Rio	Diamond Multimedia	digital player	500,000
iPod Classic	Apple	digital player	14,500,000
Zune	Microsoft	digital player	3,200,000

A student is writing a research paper on the evolution of portable music technology during the late 1990s and early 2000s. The student wants to know the approximate number of units sold worldwide of the Zune, manufactured by Microsoft. The student finds that the number sold was approximately _____

Which choice most effectively uses data from the table to complete the statement?

- A. 14,500,000 units.
- B. 3,200,000 units.
- C. 2,200,000 units.
- D. 500,000 units.

Question 12 Skill: Evidence - Data / Graph / Table | Difficulty: Medium

GRAPH TO INSERT

Billions of Barrels of Oil Produced in 2000 and 2020

Country	2000	2020
Norway	1.14	0.66
Brazil	0.47	1.10
Saudi Arabia	3.12	3.98
Canada	0.95	1.83

A student is researching global oil production. The student learns that the total amount of oil produced worldwide increased from 27.00 billion barrels in 2000 to 33.40 billion barrels in 2020. Considering a table with information about four countries, the student notes that the global increase in oil production occurred even though _____

Which choice most effectively uses data from the table to complete the statement?

- A. each of the four countries produced less than 1 billion barrels in 2000.
- B. Norway produced less oil in 2020 than in 2000.

- C. Canada produced more oil in 2020 than in 2000.
- D. each of the four countries produced more than 1 billion barrels in 2020.

Question 13 Skill: Quotation Illustrates Claim | Difficulty: Medium

Great Expectations is an 1861 novel by Charles Dickens set in early Victorian England. In the novel, the young protagonist Pip attends a grand dinner at a wealthy estate, which the narrator attributes to Pip's sense of self-conscious ambition: _____

Which quotation from *Great Expectations* best illustrates the claim?

- A. "There was a gay fiction among us that we were constantly enjoying ourselves, and a skeleton truth that we never did."
- B. "In the little world in which children have their existence, there is nothing so finely perceived as injustice."
- C. "Pip had a restless aspiration that made even the act of sitting at a fine table feel both thrilling and agonizing, for he could not forget the forge from which he came."
- D. "It is a most miserable thing to feel ashamed of home, and yet the feeling was upon me."

Question 14 Skill: Evidence - Support / Weaken | Difficulty: Medium

Melbourne, Australia, has installed solar-powered smart bins—compacting waste containers equipped with fill-level sensors—across 75% of its central business district to reduce collection costs and improve urban cleanliness. Investigating the impact of such technology on municipal waste management efficiency in mid-sized European cities, Dr. Elena Voss et al. hypothesized that smart bins reduce collection frequency more effectively than they reduce overall waste volume. To evaluate this, the researchers tracked collection data and waste tonnage across twelve pilot sites, using a Municipal Waste Efficiency Index (MWEI), on which a high score corresponds to high efficiency.

Which finding, if true, would most directly support the researchers' hypothesis?

- A. Pilot sites with a high percentage of smart bins and low traditional bin coverage had higher average MWEI scores than did pilot sites with lower smart bin coverage but similar total bin counts.
- B. Pilot sites in temperate climates with high smart bin density had lower average MWEI scores than did pilot sites in similar climates with lower smart bin density.
- C. Pilot sites with both high smart bin coverage and high traditional bin coverage had similar MWEI scores, whereas pilot sites with low coverage of both types had widely differing MWEI scores.
- D. Pilot sites where smart bins were introduced showed significantly reduced collection trips but only marginal decreases in total waste tonnage, compared to control sites without smart bins.

Question 15 Skill: Inference / Complete the Text | Difficulty: Medium

All commercial batteries contain varying combinations of metals, electrolytes, and charge-carrying compounds. However, lithium-ion batteries, commonly used in smartphones, contain a higher proportion of cobalt (at least 15%) than do nickel-metal hydride batteries, which is responsible for their greater energy density. Unlike lithium-ion batteries, nickel-metal hydride batteries have a crystalline structure that relies on a different electrode chemistry. Nickel-metal hydride batteries have two main variants: the standard type, often used in

hybrid vehicles, and the low-self-discharge type, which retains more charge over time and is commonly used in household electronics. Thus, batteries used to manufacture _____

Which choice most logically completes the text?

- A. smartphones will have greater energy density than batteries used in household electronics will.
- B. hybrid vehicles will have a cobalt proportion lower than 15%, while batteries used for smartphones will not.
- C. smartphones will have a crystalline electrode structure.
- D. household electronics will likely have a similar electrode chemistry to batteries used to manufacture smartphones.

Question 16 *Skill: Grammar / Conventions | Difficulty: Medium*

In Austin, Texas, protected cycling lanes like the Lance Armstrong Bikeway—which is about 6.2 miles long—have become an increasingly popular means of travel for both commuters and recreational cyclists alike. Moreover, Daniela _____ has identified expanded bike infrastructure as one key strategy to reduce vehicle congestion, noting the city’s mild winters and relatively level terrain.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. Gutierrez-Pham a transportation planner and cycling advocate
- B. Gutierrez-Pham, a transportation planner and cycling advocate,
- C. Gutierrez-Pham a transportation planner and cycling advocate,
- D. Gutierrez-Pham, a transportation planner and cycling advocate

Question 17 *Skill: Grammar / Conventions | Difficulty: Medium*

Stonehenge in Wiltshire and the Ring of Brodgar in Orkney are two of several British Isles _____ enormous stone circles. A distinct pattern of lichen growth covers the standing stones at Stonehenge, while a different pattern of lichen covers the stones at Brodgar.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. sites, that have their own historic stone circles, and an
- B. sites that have their own historic stone circles, an
- C. sites, that have their own historic stone circles and an
- D. sites that have their own historic stone circles. An

Question 18 *Skill: Grammar / Conventions | Difficulty: Medium*

With one station in Alaska and another in Norway, the seismic monitoring sites that form the Global Seismographic Network—or GSN, as geophysicists like Dr. Rhonda Butler call it—number nearly one hundred and fifty _____ across the globe, the seismometers record earthquake activity and other subsurface phenomena.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. stations. Located
- B. stations, located
- C. stations, which are located
- D. stations located

Question 19 *Skill: Grammar / Conventions | Difficulty: Medium*

“Catalyst,” like other three-syllable words that pair exclusively with one other word (e.g., “minister” and “sinister”), _____ with “analyst” to form a closed rhyming pair.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. couple
- B. are coupled
- C. couples
- D. have coupled

Question 20 Skill: Grammar / Conventions | Difficulty: Medium

As varied as their contributions may seem, connections exist between Rosalind Franklin, British biophysicist of the Cambridge _____ Flemish cartographer of the Mercator tradition; and Ada Lovelace, British mathematician of the Victorian era.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. School, Gerardus Mercator,
- B. School; Gerardus Mercator,
- C. School, Gerardus Mercator;
- D. School Gerardus Mercator,

Question 21 Skill: Grammar / Conventions | Difficulty: Medium

Bioluminescence, the production of light by living organisms, is a biological _____ widely understood to serve functions ranging from predator avoidance to mate attraction in a variety of deep-sea organisms—including jellyfish, anglerfish, and lanternfish—the phenomenon provides similar advantages, albeit through different biochemical pathways, to certain terrestrial insects, such as fireflies.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. mechanism
- B. mechanism,
- C. mechanism that is
- D. mechanism. Now

Question 22 Skill: Transitions | Difficulty: Medium

In 1962, Rachel Carson’s *Silent Spring* became the first major work by an American environmental scientist to be serialized in a mainstream magazine, appearing in *The New Yorker*. _____ many other works by environmental writers were published in prominent periodicals, including Barry Commoner’s *The Closing Circle* in 1971 and Bill McKibben’s *The End of Nature* in 1989.

Which choice completes the text with the most logical transition?

- A. That is,
- B. Previously,
- C. For example,
- D. Later,

Question 23 Skill: Transitions | Difficulty: Medium

The element neon is exceptionally stable, consistently ranking as one of the least reactive elements on the periodic table (neon is listed as entirely inert in most standard references). By contrast, the reactivity of the alkali metal sodium varies considerably. Indeed, when exposed to water between March and August 2023, sodium samples reacted with an astonishing increase in vigor; _____ their explosive tendency rose from moderate to extreme.

Which choice completes the text with the most logical transition?

- A. in sum,
- B. in turn,
- C. specifically,
- D. for example,

Question 24 *Skill: Transitions | Difficulty: Medium*

Generally, migratory birds travel longer distances than resident birds. The Arctic tern, _____ completes a round-trip migration of roughly 44,000 miles annually, while a typical house sparrow remains within a few miles of its nesting site throughout its life.

Which choice completes the text with the most logical transition?

- A. however,
- B. meanwhile,
- C. for example,
- D. additionally,

Question 25 *Skill: Rhetorical Synthesis | Difficulty: Medium*

While researching a topic, a student has taken the following notes:

- The Cliff Palace in southwestern Colorado is a former Ancestral Puebloan dwelling, constructed beneath a rock overhang on the side of a canyon.
- Taos Pueblo in north-central New Mexico is another former Ancestral Puebloan settlement.
- It was constructed on an open plain and has been continuously inhabited since approximately 1000 CE.
- The Ancestral Puebloan civilization was the precursor of modern Pueblo tribal nations.

The student wants to contrast Cliff Palace and Taos Pueblo. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Cliff Palace was built beneath a rock overhang on the side of a canyon, whereas Taos Pueblo was built on an open plain.
- B. In contrast to Cliff Palace, Taos Pueblo was built beneath a rock overhang on the side of a canyon.
- C. Although they were once occupied by the Ancestral Puebloan civilization, Cliff Palace and Taos Pueblo are no longer inhabited.
- D. Unlike Cliff Palace, Taos Pueblo was constructed by the Ancestral Puebloan civilization, the precursor of modern Pueblo tribal nations.

Question 26 *Skill: Rhetorical Synthesis | Difficulty: Medium*

While researching a topic, a student has taken the following notes:

- The Booker Prize is an international literary award administered by the Booker Prize Foundation in London, England.

- Novelist Hilary Mantel of the UK won a Booker Prize in 2009.
- Mantel's winning novel was entitled *Wolf Hall*.
- Novelist Douglas Stuart of Scotland won a Booker Prize in 2020.
- Stuart's winning novel was entitled *Shuggie Bain*.

The student wants to emphasize a similarity between the two novels. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Both *Wolf Hall* and *Shuggie Bain* are Booker Prize-winning novels.
- B. Mantel, a novelist, won a Booker Prize in 2009 for *Wolf Hall*.
- C. The Booker Prize Foundation in London has awarded Booker Prizes to both British and Scottish novelists.
- D. In contrast to *Wolf Hall*, a novel by Mantel, *Shuggie Bain* is a novel by Stuart.

Question 27 Skill: Rhetorical Synthesis | Difficulty: Medium

While researching a topic, a student has taken the following notes:

- Greenhouses and other controlled-environment facilities use heating systems to maintain optimal growing temperatures.
- A common heating element is a gas-fired boiler, which burns natural gas to produce heat.
- When a boiler absorbs fuel energy, it undergoes combustion, converting chemical energy to thermal energy.
- When the system cools, excess heat dissipates through radiation and convection.
- The heating element propane is a hydrocarbon fuel.
- Hydrocarbons are composed of the elements hydrogen and carbon.

The student wants to describe the role of heating systems in greenhouses. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Greenhouses can produce heat and maintain temperatures through combustion and heat dissipation.
 - B. The heating element propane, used in greenhouses, is composed of hydrogen and carbon.
 - C. Used in greenhouses, boilers and other heating elements can convert chemical energy to thermal energy and back.
 - D. Greenhouses use heating systems like gas-fired boilers to maintain optimal growing temperatures by burning fuel.
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Section 1: Reading and Writing

Module 2 - Easier Path (27 Questions)

Time: 32 minutes

Question 1 *Skill: Words in Context | Difficulty: Easy*

Although West African literature is highly diverse in its themes and forms, it is also characterized by considerable thematic _____. scholars have drawn connections linking works by writers such as Chinua Achebe and Wole Soyinka to the oral storytelling traditions of the Yoruba and Igbo peoples.

Which choice completes the text with the most logical and precise word or phrase?

A. subtlety

- B. sophistication
- C. continuity
- D. innovation

Question 2 Skill: Words in Context | Difficulty: Easy

When discussing artistic movements in France, critics regularly characterize the paintings of Claude Monet as especially representative of Impressionism. The reasons for this characterization may seem _____, but linking Monet exclusively with Impressionism risks disregarding the complexities in his later work that do not neatly conform to the conventions of this artistic movement.

Which choice completes the text with the most logical and precise word or phrase?

- A. incongruous
- B. pretentious
- C. provocative
- D. irrefutable

Question 3 Skill: Words in Context | Difficulty: Easy

The following text is from a 1960 translation of Fyodor Dostoevsky’s 1866 novel *Crime and Punishment* (translated by Constance Garnett). In the text, Raskolnikov has climbed to the top of a bridge overlooking the Neva River. And having reached the railing he had to confess the evening was fading, gathering itself into the deeper shadow. Yes, the last glow was unravelling, spreading across the horizon, and there below, the river, already dark, was manifest in silvery flashes of reflected lamplight, darting and dissolving, ever more faint and fitful, and destined no sooner lit to be swallowed up.

As used in the text, what does the word “manifest” most nearly mean?

- A. Situated
- B. Perceptible
- C. Realized
- D. Dwindling

Question 4 Skill: Words in Context | Difficulty: Easy

The sweeping archways and ornamental tilework of the Taj Mahal in Agra, India, exemplify the Mughal contribution to South Asian architecture, an influence that is _____ throughout the subcontinent—visitors touring Humayun’s Tomb in Delhi, for instance, can readily identify how its layout draws inspiration from Persian garden-tomb traditions.

Which choice completes the text with the most logical and precise word or phrase?

- A. corroborated
- B. understated
- C. disputed
- D. palpable

Question 5 Skill: Overall Structure | Difficulty: Easy

In 2018, Tomoko Hayashi and colleagues published a study concluding that ocean warming has a significant effect on the migration patterns of *Thunnus alalunga*, a species of tuna. However, Hayashi and colleagues’ study relied on a tracking sample of only 8 fish. In a 2023

review of various scientists' conclusions about the impacts of ocean warming on pelagic fish behavior, Dr. Marcus Reid and colleagues caution that relying on such a small sample size can increase the potential for biased analysis. *Such analysis, in turn, can contribute to reports of exaggerated effects.*

Which choice best describes the function of the underlined sentence in the text as a whole?

- A. It summarizes a shift in scientists' understanding of how *Thunnus alalunga* has responded to ocean warming.
- B. It elaborates on a potential consequence of Hayashi and colleagues' reliance on a relatively small sample size.
- C. It emphasizes the magnitude of the effect reported by Hayashi and colleagues of ocean warming on *Thunnus alalunga*.
- D. It counters the objection of Reid and colleagues to studies that rely on relatively small sample sizes.

Question 6 Skill: Overall Structure | Difficulty: Easy

The following text is from a 1922 translation of Guy de Maupassant's 1885 short story "The Necklace." The narrator is describing Madame Loisel, a clerk's wife in Paris. *For the last ten years she had walked the same path to the market each morning, and had greeted the same shopkeepers at the same hour, and nearly at the same stall. Every morning, after haggling over the price of vegetables, she carried her basket home, like a soldier returning from an exhausting campaign, and sat heavily in her chair, always feeling defeated, as though she were expecting some reprieve from a life she had never chosen.*

Taken together, the three underlined portions most clearly serve which function in the text as a whole?

- A. They illustrate an unexplained nervous tension that Loisel experiences as she begins each day.
- B. They suggest Loisel's weariness with being trapped in a life that has become oppressively routine.
- C. They reveal that Loisel is unwaveringly precise in her daily routine because of the strict expectations of her husband.
- D. They convey Loisel's growing apprehensiveness about having failed to complete an important errand for the first time in her long marriage.

Question 7 Skill: Main Purpose | Difficulty: Easy

Welwitschia mirabilis is a species of plant found exclusively in the Namib Desert, one of the oldest deserts on Earth. Members of this species exhibit an extraordinary range of adaptations, with some individuals developing extensive root networks to tap underground water sources while others survive primarily on coastal fog moisture. All specimens of *Welwitschia* descended from a single ancestral gymnosperm lineage that arrived in the region around 100 million years ago. The lineage's descendants diversified into distinct survival strategies as they adapted to the extreme aridity of the Namib.

Which choice best states the main purpose of the text?

- A. To advance the claim that all plants in the Namib Desert are part of the *Welwitschia* species and list their possible ancestors
- B. To indicate the specific ancestral gymnosperm of all desert plants and explain why the plants have such varied physical characteristics
- C. To describe the *Welwitschia* species and explain how the plant

- family developed such varied survival strategies
- D. To describe the specific habitat where *Welwitschia mirabilis* is found and identify other plants that share a common ancestor with it

Question 8 Skill: Cross-Text | Difficulty: Easy

Text 1 One challenge faced by researchers studying global food security is that countries may define food insecurity differently. Many countries measure food insecurity based on caloric intake per capita. However, countries do not all use the same thresholds; for example, Ethiopia uses a minimum daily intake of 1,800 calories, while Germany uses 2,100. Other countries use a combination of caloric intake and nutritional diversity measures. This variation makes it difficult for researchers to compare aspects of food insecurity across different countries.

Text 2 Recently, a consortium of five international health organizations developed standardized global classifications for levels of food insecurity. The consortium developed a new framework called the “Integrated Food Security Phase Classification.” This new framework establishes global criteria used to define five phases of food insecurity (minimal, stressed, crisis, emergency, and famine) and allows researchers to better compare global food security conditions.

Based on the texts, how would the author of Text 2 most likely respond to the problem presented in Text 1?

- A. By suggesting that researchers focus on topics besides food security
- B. By noting that a possible solution to the problem is available
- C. By recommending that a specific institution should further investigate the problem
- D. By supplying additional ways in which food security research is difficult

Question 9 Skill: Inference / Complete the Text | Difficulty: Easy

Dr. Amara Okafor and Dr. Jun Li’s research on the fiscal administration of Emperor Akbar (who ruled the Mughal Empire from 1556 to 1605) builds on other studies of South Asian early modern state finance, including Habib’s and Richards’s work on the relationship between land revenue systems and political stability. But Okafor and Li’s unique contribution is their reconstruction of the earliest surviving complete set of provincial revenue records for any Mughal ruler, which demonstrate that Akbar’s periodic treasury shortfalls were caused by disruptions in provincial remittance cycles, not systemic administrative corruption.

What does the text most strongly suggest about the value of Okafor and Li’s research to their discipline?

- A. Okafor and Li’s research uses newly compiled evidence to eliminate one possible reason Akbar faced treasury shortfalls and to affirm another.
- B. Okafor and Li’s research revealed the role of Akbar’s treasury shortfalls in exacerbating provincial unrest.
- C. Okafor and Li’s research builds on earlier work about Mughal state finance by Habib and Richards and corrects errors in that earlier work.
- D. Okafor and Li’s research presented a novel body of evidence supporting Habib and Richards’s hypothesis regarding why Akbar faced recurring fiscal problems.

Question 10 Skill: Evidence - Support / Weaken | Difficulty: Easy

Dr. Maria Volkov and Dr. James Osei investigated the role of scent production in pollination among orchid species. They found that during controlled experiments, 42% of the tested species produced detectable scent compounds, which ranged from sweet floral notes in *Vanilla planifolia* to musty odors in *Dracula vampira*. A second research team has claimed that orchids use these scents primarily to attract specific pollinator species.

Which finding, if true, would most directly challenge the second team's claim?

- A. Among the orchid species that produced scent in response to pollinator presence, no individuals produced scent before pollinators arrived, but the vast majority produced scent once pollinators appeared.
- B. Most of the orchid species that were found to not produce scent in response to pollinator presence have been observed producing scent during encounters with herbivorous insects.
- C. Orchids that were found to produce scent in pollinator experiments are typically self-pollinating and were tested in isolation.
- D. The scents orchids produced in response to pollinators had acoustic-frequency characteristics that would make them undetectable to bees, butterflies, or moths, the most frequent pollinators of orchids.

Question 11 Skill: Evidence - Support / Weaken | Difficulty: Easy

Nike's expansion into wearable fitness technology in 2012 is a quintessential instance of brand extension—the company leveraged its brand recognition as an athletic apparel provider to enter a product category where it had not previously competed. An outstanding question is whether perceived category similarity predicts consumers' likelihood of purchasing brand extensions. To answer this question, Dr. Rajan Patel et al. identified 25 extended-brand pairs (e.g., the same brand of shoes and smartwatches) in 48 weeks of purchases by approximately 50,000 households and, for each pair, calculated the change in probability of a brand in one category being purchased if the same brand was purchased in the other category.

Based on the text, which potential study design would be most likely to produce evidence that would enable Patel et al. to answer their research question?

- A. Have a representative sample of the households rate the similarity of the product categories in each extended-brand pair, then determine how, if at all, those ratings correlate with the change in probability that the team calculated for each pair.
- B. Poll a representative sample of the households to determine the degree of brand recognition for each brand in the extended-brand pairs, then determine how, if at all, the degree of brand recognition correlates with the average cost of each product in the pairs.
- C. Have a representative sample of the households rate the similarity of one product in each extended-brand pair to other products in the same category, then determine how, if at all, those ratings correlate with the change in probability that the team calculated for each pair.
- D. Poll a representative sample of the households to determine the degree of brand recognition for each brand in the extended-brand

pairs, then determine how, if at all, the degree of brand recognition correlates with the frequency with which a different group of households purchased at least one product of that brand.

Question 12 Skill: Evidence - Data / Graph / Table | Difficulty: Easy

GRAPH TO INSERT

[Graph: Bar chart titled “Correlation between Model-Predicted and Student-Reported Satisfaction Ratings, by Subject Area.” X-axis shows two subject areas: STEM and Humanities. Y-axis shows Correlation from 0 to 0.5. Three bars per subject area represent students P1 (light gray), P4 (white), and P3 (dark gray). For STEM: P1 ~ 0.15, P4 ~ 0.35, P3 ~ 0.20. For Humanities: P1 ~ 0.20, P4 ~ 0.42, P3 ~ 0.25.]

Education researcher Dr. Kenji Tanaka and colleagues developed a computational model to predict how satisfied a student will be with a particular course on a scale from 1 (not at all) to 5 (very much). They then recruited participants to use the same scale to rate several sets of courses in various subject areas and calculated the correlation between the ratings predicted by the model and those reported by the participants. Assuming participant P4 gave equal ratings to the STEM and Humanities courses, the data in the graph indicate the model predicted that _____

Which choice most effectively uses data from the graph to complete the example?

- A. P4’s ratings for STEM and Humanities courses would differ from one another.
- B. P4 would derive less satisfaction from STEM courses than from Humanities courses.
- C. P4’s ratings for STEM and Humanities courses would equal one another.
- D. P4 would derive more satisfaction from STEM courses than from Humanities courses.

Question 13 Skill: Inference / Complete the Text | Difficulty: Easy

To understand brand loyalty among subscribers to specific streaming platforms, Dr. Nina Johansson and colleagues conducted a survey using 42 statements as proxy indicators of subscriber satisfaction with their streaming experience. The statements were categorized by topic—e.g., statement 15, “I look forward to new content releases on this platform,” was categorized as anticipatory engagement—and respondents, all of whom were from South Korea (which is characterized as having a highly competitive streaming market), rated the importance of each statement to their experience. Researchers found that respondents placed high importance on content variety and recommendation accuracy but low importance on interface design, but the researchers cautioned against applying the findings to subscribers generally, suggesting that _____

Which choice most logically completes the text?

- A. additional research is needed with participants from countries of varying levels of streaming market competition to determine whether content variety and recommendations are more important than interface design to subscribers broadly.
- B. subscribers in countries with less competitive streaming markets than that of South Korea are likely to regard interface design as relatively more important than content variety and recommendation accuracy.
- C. streaming platforms that operate in a variety of countries are more likely to increase subscriber retention if they invest in content

variety and allow subscribers to customize recommendations than if they redesign their interfaces.

- D. the greater importance assigned to content variety and recommendations than to interface design may not be observed if the survey were to be given to a larger group of South Korean participants.

Question 14 Skill: Inference / Complete the Text | Difficulty: Easy

Dr. Rachel Simmons et al. tracked temperature-driven changes in the population of *Daphnia magna* (a freshwater crustacean, or zooplankton, that requires two host organisms throughout its life cycle), *Bosmina longirostris* (a directly reproducing zooplankton, which requires only one environment), and 65 other zooplankton taxa found in twelve lake systems. Zooplankton with complex life cycles are typically shielded from thermal stress by their host organisms, whereas directly reproducing zooplankton are exposed to external temperature conditions. However, Simmons et al. found that two-host zooplankton abundance decreased as water temperatures rose, whereas directly reproducing zooplankton abundance was largely stable, suggesting that _____

Which choice most logically completes the text?

- A. zooplankton primarily transmitted through host organisms were less dependent on host species adversely affected by warming temperatures than were zooplankton that use other reproductive strategies.
- B. as the number of host species involved in a zooplankton's life cycle increases, the zooplankton is better protected against rising temperatures.
- C. directly reproducing zooplankton identified in the study were more likely to use reproductive strategies that shield them from warming temperatures than were two-host zooplankton.
- D. any advantages that the life-cycle strategy used by two-host zooplankton may have conferred did not fully offset the negative effects of other temperature-driven factors on their abundance.

Question 15 Skill: Inference / Complete the Text | Difficulty: Easy

Prolonged exposure to artificial light at night (light from human sources like streetlamps or buildings) can affect wildlife, as Dr. Kevin Park and colleagues found in a 2015 study of urban-dwelling barn owls. Researchers conducted a meta-analysis of studies of how artificial light affects nocturnal animals and found that, for every study, relevant traits or behaviors of the animals were observably different between the light-exposed group and the otherwise similar but unexposed group. Although, on average, studies of birds showed larger differences than studies of mammals did, for every class of animals examined, there were individual studies showing differences well above the average for birds. Therefore, the results of the meta-analysis suggest that _____

Which choice most logically completes the text?

- A. some studies of mammals found larger effects of exposure to artificial light at night than some studies of birds did.
- B. the studies in the meta-analysis that examined birds were more likely than those that examined mammals to specify whether the observed effects were detrimental.
- C. the differences that studies attribute to exposure to artificial light at night are likely to be more pronounced for mammals than they are for birds.

- D. the difference found in the study conducted by Dr. Kevin Park and colleagues was likely larger than the average difference for studies of barn owls included in the meta-analysis.

Question 16 *Skill: Grammar / Conventions | Difficulty: Easy*

The Ottoman Empire, which dominated the eastern Mediterranean from around 1299 CE to 1922 CE, prospered from trade with neighboring civilizations: its merchants exported textiles, ceramics, and spices and _____ other luxury goods, such as silk, precious metals, and gemstones, which were then sold in local bazaars.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. to import
- B. imported
- C. importing
- D. having imported

Question 17 *Skill: Grammar / Conventions | Difficulty: Easy*

In her work, Sami-heritage artist Britta Marakatt-Labba uses thread, fabric, and other textile mediums to explore Indigenous identity and Arctic landscapes. The National Museum of Norway counts pieces by Marakatt-Labba in _____ impressive collection of Nordic art.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. their
- B. it's
- C. his
- D. its

Question 18 *Skill: Grammar / Conventions | Difficulty: Easy*

On July 14, 2015, NASA's New Horizons spacecraft detected a remarkable and previously unknown geological feature on Pluto, located in the Tombaugh Regio, 3.26 billion miles from _____ observation of a vast nitrogen-ice plain, an extraordinarily smooth expanse that typically results from ongoing geological resurfacing of a dwarf planet's crust.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. Earth. The
- B. Earth, it was the
- C. Earth: the
- D. Earth, the

Question 19 *Skill: Grammar / Conventions | Difficulty: Easy*

Between October 15 and November 25 each year, debris shed by comet 55P/Tempel-Tuttle enters Earth's atmosphere at a velocity of 71 kilometers per second. Relatively few of these particles are likely to reach Earth's _____ since the meteoroids' friction with the atmosphere causes the vast majority to burn up in the mesosphere, producing the Leonid meteor shower.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. surface; however,
- B. surface, however,
- C. surface. However,
- D. surface, however;

Question 20 *Skill: Grammar / Conventions | Difficulty: Easy*

In a chemical reaction, the quantity known as molar mass helps convert between the mass of the reactants and the mass of the product. The liquid _____ have molar masses of 180.16 and 32.04 g/mol, respectively.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. compounds, glucose and methanol
- B. compounds, glucose and methanol,
- C. compounds glucose and methanol
- D. compounds glucose and methanol,

Question 21 *Skill: Grammar / Conventions | Difficulty: Easy*

The philosophy of the Montessori method is rooted in several central _____ by Italian physician Maria Montessori, who sought to parallel the natural developmental stages of children, the method emphasizes learning from hands-on materials and self-directed activity and introduces students as young as two to practical life exercises such as pouring and sorting.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. tenets developed
- B. tenets, developed
- C. tenets. Developed
- D. tenets that, developed

Question 22 *Skill: Transitions | Difficulty: Easy*

Margaret Bourke-White and Gordon Parks have both received Lifetime Achievement Awards from the International Center of Photography, though for different contributions. Bourke-White was honored for her pioneering work in photojournalism; Parks, _____ was recognized for his contributions as both a photographer and filmmaker.

Which choice completes the text with the most logical transition?

- A. as a result,
- B. for instance,
- C. in other words,
- D. by contrast,

Question 23 *Skill: Transitions | Difficulty: Easy*

In a 2021 analysis of 150 terms, researchers found a broad pattern of frequency-dependent change: words used less often in everyday speech saw a faster rate of semantic drift—_____ the rate at which a word's meaning will shift over time. Nouns (e.g., "courage") saw the largest effect; verbs (e.g., "gather"), meanwhile, saw the smallest.

Which choice completes the text with the most logical transition?

- A. likewise,

- B. that is,
- C. for example,
- D. in addition,

Question 24 Skill: Transitions | Difficulty: Easy

At previous World Fairs, in contrast to exhibits showcasing international technological achievements, regional folk crafts were displayed as limited-run demonstrations. _____ while the internationally recognized art of glassblowing became a permanent exhibition after its initial demonstration, ikebana, a Japanese flower-arranging art popular mainly in East Asia, only appeared at the 1970 Osaka Expo.

Which choice completes the text with the most logical transition?

- A. Moreover,
- B. However,
- C. Granted,
- D. Accordingly,

Question 25 Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

- Composer Toru Takemitsu was a Japanese musician and citizen of Tokyo.
- His compositions blend Western orchestral music with elements of traditional Japanese aesthetics.
- Takemitsu's orchestral piece *November Steps* features a biwa, a traditional Japanese lute.
- Takemitsu's orchestral piece *A Flock Descends into the Pentagonal Garden* incorporates a shakuhachi, a traditional Japanese bamboo flute.

The student wants to emphasize a difference between the two compositions. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. While both compositions integrate traditional Japanese instruments, *November Steps* does so by featuring a biwa and *A Flock Descends into the Pentagonal Garden* does so by incorporating a shakuhachi.
- B. *November Steps* and *A Flock Descends into the Pentagonal Garden* are two compositions written by Takemitsu, a Japanese composer.
- C. In his two compositions *November Steps* and *A Flock Descends into the Pentagonal Garden*, Takemitsu blends elements of traditional Japanese instruments, such as lutes or flutes, with Western orchestral music.
- D. Takemitsu has different approaches to blending Western orchestral music with elements of Japanese tradition, such as using Japanese lutes and flutes in his compositions.

Question 26 Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

- An oxbow lake is a U-shaped body of water formed when a river meander is cut off from the main channel.
- An oxbow lake forms when sediment deposits seal both ends of the meander, isolating it from the river.
- Oxbow lakes will gradually shrink and become wetlands over time.
- A river with high sinuosity has many meanders, and a river with

- low sinuosity has few.
- The Mississippi River in the United States has high sinuosity.

The student wants to define the term “high-sinuosity river.” Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. A high-sinuosity river is one that has many meanders, or U-shaped curves.
- B. A high-sinuosity river has U-shaped curves called meanders that will gradually change shape and shift downstream.
- C. High sinuosity is caused by the formation of oxbow lakes from river meanders over time.
- D. Over time, the many meanders in the high-sinuosity Mississippi River will change shape and migrate downstream.

Question 27 Skill: Rhetorical Synthesis | Difficulty: Easy

While researching a topic, a student has taken the following notes:

- Jupiter has at least 95 known moons, including Europa and Ganymede.
- Europa completes an orbit of Jupiter in 3.55 Earth days on average.
- Ganymede completes an orbit of Jupiter in 7.15 Earth days on average.
- German astronomer Johannes Kepler (1571–1630) described the orbits of planets as elliptical in shape rather than circular.
- Kepler published three laws of planetary motion to describe the orbits of planets around the Sun.
- Kepler’s laws also apply to the elliptical orbits of natural satellites (e.g., moons).

The student wants to contrast the orbits of the two moons. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Europa’s orbit of Jupiter is elliptical in shape and takes, on average, 3.55 Earth days, while Ganymede’s orbit of Jupiter is also elliptical and takes, on average, 7.15 Earth days.
 - B. Kepler’s three laws of planetary motion describe the orbits of planets and natural satellites, including those of Europa and Ganymede.
 - C. Europa completes its orbit of Jupiter many times faster than Ganymede does.
 - D. Jupiter’s moons, specifically Europa and Ganymede, orbit their planet in the same way the planets orbit the Sun.
-

Section 1: Reading and Writing

Module 2 - Harder Path (27 Questions)

Time: 32 minutes

Question 1 *Skill: Words in Context | Difficulty: Hard*

The Silk Road was not a single thoroughfare but a sprawling lattice of overland and maritime routes connecting East Asia, the Middle East, and the Mediterranean. Despite their geographic dispersion, the artistic traditions that emerged along these routes display a striking

thematic ____: scholars have traced recurring design motifs—such as interlocking arabesques and lotus medallions—from Tang-dynasty ceramics through Abbasid textiles to Byzantine mosaics.

Which choice completes the text with the most logical and precise word or phrase?

- A. ambiguity
- B. coherence
- C. spontaneity
- D. uniformity

Question 2 *Skill: Words in Context | Difficulty: Hard*

Art historians routinely classify Frida Kahlo's paintings as belonging to Surrealism. The basis for this classification may seem ____, given Kahlo's dream-like imagery and symbolic self-portraits, but applying the label too broadly risks obscuring the deeply personal, politically grounded dimensions of her work that resist reduction to any single movement.

Which choice completes the text with the most logical and precise word or phrase?

- A. incongruous
- B. derivative
- C. incontrovertible
- D. unsubstantiated

Question 3 *Skill: Words in Context | Difficulty: Hard*

Under European Union regulations, it is ____ for experimentalists conducting clinical drug trials to adjust dosage protocols mid-study if unexpected adverse effects arise, but they are not compelled to do so; the final decision rests with the institutional review board overseeing the trial.

Which choice completes the text with the most logical and precise word or phrase?

- A. permissible
- B. unprecedented
- C. compulsory
- D. negligible

Question 4 *Skill: Words in Context | Difficulty: Hard*

The monumental frescoes adorning the Sistine Chapel in Vatican City exemplify the Renaissance contribution to Western art, an influence that is ____ throughout European cultural institutions—visitors touring the Uffizi Gallery in Florence, for instance, can readily identify how its holdings draw upon the humanist principles championed by Michelangelo and his contemporaries.

Which choice completes the text with the most logical and precise word or phrase?

- A. understated
- B. corroborated
- C. disputed
- D. conspicuous

Question 5 *Skill: Overall Structure | Difficulty: Hard*

The following text is adapted from Joseph Conrad's 1899 novella *Heart of Darkness* (published serially in *Blackwood's Magazine*). The narrator is describing the arrival of a colonial officer at a trading outpost. The officer dismounted and surveyed the assembled porters with the deliberate indifference of a magistrate inspecting prisoners, as if to communicate: "I have administered territories from Mombasa to Leopoldville, and you should consider yourselves fortunate that I have consented to oversee this wretched station." *There are administrators who regard colonial outposts as personal dominions rather than shared enterprises, and it is widely observed that such persons inspire obedience without loyalty.*

Which choice best describes the function of the underlined sentence in the text as a whole?

- A. It explains why colonial trading outposts were difficult to administer.
- B. It characterizes the porters' reaction to the officer's arrival.
- C. It suggests that the officer's managerial style is ultimately ineffective.
- D. It illustrates the officer's condescending attitude toward the people at the outpost.

Question 6 Skill: Overall Structure | Difficulty: Hard

Toni Morrison's *Beloved*, first published in 1987, is a novel in Morrison's broader exploration of African American historical memory. Like the rest of Morrison's major works, *Beloved* has a deliberately fragmented narrative structure, but the subsequent novel *Jazz* extends many of the same themes of communal memory and identity. *Thus, the fragmentation of the narrative is only disorienting if one reads the novels as self-contained works*—Morrison's oeuvre is best understood as an interconnected meditation on history, similar to William Faulkner's Yoknapatawpha novels.

Which choice best describes the function of the underlined portion in the text as a whole?

- A. It argues that Morrison's novels are more critically acclaimed than Faulkner's Yoknapatawpha novels.
- B. It demonstrates that *Jazz* is more accessible to readers who have first read *Beloved*.
- C. It offers a framework for understanding why the narrative fragmentation of *Beloved* should not be seen as a deficiency.
- D. It presents evidence that *Beloved* and *Jazz* are considered Morrison's most structurally complex novels.

Question 7 Skill: Overall Structure | Difficulty: Hard

Argentine writer Jorge Luis Borges's literary career, which spanned the 1920s to the 1980s, evolved through distinct phases. As epitomized by the collection *Fervor de Buenos Aires*, many of Borges's early works were lyric poems rooted in the landscapes and nostalgia of his native city. *An extended period of philosophical immersion—particularly in the works of Schopenhauer, Berkeley, and the paradoxes of Zeno—later catalyzed a turn toward metaphysical fiction*, a shift whose emphasis on labyrinths, infinite libraries, and the unreliability of perception is readily apparent in Borges's celebrated collection *Ficciones*.

Which choice best describes the function of the underlined portion in the text as a whole?

- A. It presents biographical information that accounts for the transition in Borges's career that is discussed in the text.
- B. It notes why the two literary works described in the text constitute major departures from Borges's typical style.
- C. It lends support to an argument about the lyric aesthetic made in the previous sentence.
- D. It explains why the literary work discussed later in the sentence was praised by Borges's contemporaries.

Question 8 Skill: Main Idea | Difficulty: Hard

Ghanaian sculptor El Anatsui transforms public spaces using his distinctive method of weaving discarded bottle caps and aluminum fragments into monumental tapestries. In some installations, Anatsui focuses on evoking the transatlantic slave trade, as he does in his wall-sized work *Dusasa*. However, in other works, Anatsui celebrates the resourcefulness of everyday life, as he does in his shimmering depiction of a cloth merchant's stall in an Accra market. Anatsui views both approaches as ways of representing the resilience and interconnectedness of West African communities.

Which choice best states the main idea of the text?

- A. Anatsui's installation *Dusasa* has received more scholarly attention than his depiction of the cloth merchant's stall.
- B. Anatsui's installations honor both historically significant events and everyday examples of communal resourcefulness and connection.
- C. Anatsui began his artistic career by creating installations about the slave trade and later transitioned to depicting everyday market scenes.
- D. Anatsui's depiction of a cloth merchant's stall focuses more on individual enterprise than on communal bonds.

Question 9 Skill: Main Idea | Difficulty: Hard

In their analysis of corporate wellness programs, Dr. Leah Thornton et al. confirm that such programs, though they may increase employee participation in health screenings, often fail to produce measurable reductions in overall healthcare costs. This phenomenon can be explained by the concept of moral hazard as described in Kenneth Arrow's foundational work on health economics. In this case, employees who receive wellness benefits may reduce their own health-maintenance efforts—exercising less independently or eating less carefully—effectively offsetting the gains the employer intended to achieve.

Which choice best states the main idea of the text?

- A. Arrow's concept of moral hazard provides a framework for explaining the finding by Thornton et al. that wellness programs may not reduce healthcare costs as intended.
- B. The analysis by Thornton et al. reveals that higher employee participation in wellness programs is linked to lower overall healthcare spending.
- C. Although Arrow's theory of moral hazard suggests otherwise, wellness programs can succeed as cost-reduction tools provided that they include mandatory participation.
- D. Research by Thornton et al. supports Arrow's conclusion that because employees value convenience over health, it is difficult for wellness programs to reduce costs.

Question 10 Skill: Evidence - Support / Weaken | Difficulty: Hard

Dr. Suzanne Simard and colleagues investigated the role of mycorrhizal networks—underground fungal lattices connecting tree roots—in nutrient transfer between trees. They found that during controlled experiments, Douglas fir and paper birch trees connected by mycorrhizal networks transferred significant quantities of carbon between them. A second research team has claimed that such nutrient sharing occurs primarily between parent trees and their offspring as a form of kin selection.

Which finding, if true, would most directly challenge the second team’s claim?

- A. Among the tree pairs that showed high levels of nutrient transfer, most were located in forests with dense mycorrhizal networks rather than sparse ones.
- B. The tree pairs that exhibited the highest rates of nutrient transfer through mycorrhizal networks were predominantly unrelated species rather than parent-offspring pairs.
- C. Trees that transferred nutrients through mycorrhizal networks were typically older and larger than the trees that received those nutrients.
- D. The mycorrhizal networks connecting parent trees to their offspring were, on average, denser than networks connecting unrelated trees.

Question 11 *Skill: Evidence - Support / Weaken | Difficulty: Hard*

Patagonia’s expansion into food products in 2012 (Patagonia Provisions) is an instance of mission-driven brand extension—the company leveraged its environmental brand identity to enter a product category where it had not previously competed. An outstanding question is whether perceived mission alignment predicts consumers’ willingness to pay a price premium for mission-driven brand extensions. To answer this question, Dr. Yuki Sato et al. identified 30 mission-driven brand-extension pairs in 52 weeks of purchases by approximately 60,000 households and, for each pair, calculated the average price premium paid relative to competing products.

Based on the text, which potential study design would be most likely to produce evidence that would enable Sato et al. to answer their research question?

- A. Poll a representative sample of the households to determine brand awareness for each brand in the extension pairs, then determine how, if at all, brand awareness correlates with the average product cost.
- B. Have a representative sample of the households rate the quality of products in each extension pair, then determine how, if at all, quality ratings correlate with household income.
- C. Poll a representative sample of the households to determine purchase frequency for each brand, then determine how, if at all, frequency correlates with price paid.
- D. Have a representative sample of the households rate the degree of mission alignment for each brand-extension pair, then determine how, if at all, those ratings correlate with the price premium the team calculated for each pair.

Question 12 *Skill: Evidence - Data / Graph / Table | Difficulty: Hard*

GRAPH TO INSERT

[Graph: Bar chart titled “Correlation between Model-Predicted and Student-Reported Engagement Ratings, by Course Modality.” X-axis shows two modalities: In-Person and Online. Y-axis shows Correlation

from 0 to 0.6. Three bars per modality represent students P2 (light gray), P3 (white), and P5 (dark gray). For In-Person: P2 ~ 0.18, P3 ~ 0.40, P5 ~ 0.28. For Online: P2 ~ 0.25, P3 ~ 0.52, P5 ~ 0.33.]

Education researcher Dr. Anika Larsen and colleagues developed a computational model to predict how engaged a student will be with a particular course on a scale from 1 (not at all) to 5 (very much). They then recruited participants to use the same scale to rate several sets of courses delivered through different modalities and calculated the correlation between the ratings predicted by the model and those reported by the participants. Assuming participant P3 reported identical engagement scores for In-Person and Online courses, the data in the graph indicate the model predicted that _____

Which choice most effectively uses data from the graph to complete the example?

- A. P3 would be more engaged in Online courses than in In-Person courses.
- B. P3's engagement ratings for In-Person and Online courses would equal one another.
- C. P3's engagement ratings for In-Person and Online courses would differ from one another.
- D. P3 would be more engaged in In-Person courses than in Online courses.

Question 13 Skill: Inference / Complete the Text | Difficulty: Hard

To investigate audience reception of independent cinema, Dr. Claire Fontaine and colleagues conducted a survey using 38 statements as proxy indicators of viewer satisfaction. The statements were categorized by topic—e.g., statement 22, “I would recommend this film to someone with different tastes,” was categorized as evangelistic enthusiasm—and respondents, all of whom were from France (which is characterized as having a highly developed art-house film culture), rated the importance of each statement. Researchers found that respondents placed high importance on directorial vision and narrative ambiguity but low importance on production quality, but the researchers cautioned against applying the findings to viewers generally, suggesting that _____

Which choice most logically completes the text?

- A. additional research is needed with participants from countries of varying levels of art-house film culture to determine whether directorial vision and narrative ambiguity are more important than production quality to audiences broadly.
- B. viewers in countries with less developed art-house traditions than that of France are likely to regard production quality as relatively more important than directorial vision and narrative ambiguity.
- C. independent filmmakers who distribute their films in a variety of countries are more likely to increase viewership if they prioritize directorial vision over production quality.
- D. the greater importance assigned to directorial vision and narrative ambiguity than to production quality may not be observed if the survey were given to a larger group of French participants.

Question 14 Skill: Inference / Complete the Text | Difficulty: Hard

Dr. Tomasz Kowalski et al. tracked salinity-driven changes in the population of *Artemia franciscana* (a brine shrimp, or branchiopod, that requires two host environments throughout its life cycle), *Moina macrocopa* (a directly reproducing branchiopod, which requires only

one environment), and 48 other branchiopod taxa found in fourteen salt-lake systems. Branchiopods with complex life cycles are typically shielded from osmotic stress by their host environments, whereas directly reproducing branchiopods are exposed to external salinity conditions. However, Kowalski et al. found that two-host branchiopod abundance decreased as salinity increased, whereas directly reproducing branchiopod abundance was largely stable, suggesting that _____

Which choice most logically completes the text?

- A. branchiopods primarily sheltered by host environments were less dependent on host species adversely affected by rising salinity than were branchiopods that use other reproductive strategies.
- B. as the number of host environments involved in a branchiopod's life cycle increases, the branchiopod is better protected against rising salinity.
- C. directly reproducing branchiopods identified in the study were more likely to use reproductive strategies that shield them from rising salinity than were two-host branchiopods.
- D. any advantages that the life-cycle strategy used by two-host branchiopods may have conferred did not fully offset the negative effects of other salinity-driven factors on their abundance.

Question 15 Skill: Inference / Complete the Text | Difficulty: Hard

Prolonged exposure to ocean acidification (a decrease in seawater pH caused by dissolved carbon dioxide) can affect marine organisms, as Dr. Helena Strand and colleagues found in a 2019 study of tropical coral reefs. Researchers conducted a meta-analysis of studies of how acidification affects calcifying organisms and found that, for every study, relevant traits of the organisms were observably different between the acidified group and the otherwise similar but control group. Although, on average, studies of corals showed larger differences than studies of sponges did, for every phylum of organisms examined, there were individual studies showing differences well above the average for corals. Therefore, the results of the meta-analysis suggest that _____

Which choice most logically completes the text?

- A. the studies in the meta-analysis that examined corals were more likely than those that examined sponges to specify whether the observed effects were detrimental.
- B. some studies of sponges found larger effects of exposure to ocean acidification than some studies of corals did.
- C. the differences that studies attribute to ocean acidification are likely to be more pronounced for sponges than they are for corals.
- D. the difference found by Dr. Strand and colleagues was likely larger than the average difference for coral studies included in the meta-analysis.

Question 16 Skill: Grammar / Conventions | Difficulty: Hard

The Library of Congress, which has served as the de facto national library of the United States since its founding in 1800, expanded its collections through purchases of historical _____ its merchants acquired rare manuscripts, early printed books, and cartographic materials, which were then cataloged and preserved for public access.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. archives, and
- B. archives and
- C. archives: its curators acquired
- D. archives, its curators acquired

Question 17 Skill: Grammar / Conventions | Difficulty: Hard

Roger Angell’s reputation as one of the greatest baseball writers of all _____ bolstered by dozens of *New Yorker* essays in which he demonstrated a deep affection for and knowledge of the game, has perhaps overshadowed his remarkable career as a fiction editor: during his decades-long tenure at the *New Yorker*, Angell helped shape the work of John Updike, Ann Beattie, and other contributors of note.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. time was
- B. time, was
- C. time had been
- D. time,

Question 18 Skill: Grammar / Conventions | Difficulty: Hard

Before the ratification of the 19th Amendment in 1920, many US territories extended voting rights to women, in part to spur westward migration that would help these territories meet population thresholds for statehood. The Territory of Wyoming approved women’s suffrage in 1869 before attaining statehood in _____ while eastern states, lacking the same incentive, did not follow suit until after the turn of the century.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. 1890, for example,
- B. 1890; for example,
- C. 1890, for example;
- D. 1890 for example,

Question 19 Skill: Grammar / Conventions | Difficulty: Hard

The legacy of the Spanish Empire, which once controlled portions of five continents, is evident in Spanish-speaking Panama, one of many places that reveal their imperial history in their language. Contrast Panama with the Netherlands, which ceased to be part of the empire in _____ the latter’s connection to the empire is so attenuated that Spanish is seldom spoken there today.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. 1648 and
- B. 1648:
- C. 1648,
- D. 1648

Question 20 Skill: Grammar / Conventions | Difficulty: Hard

In economics, mustard seeds are considered a commodity because they are essentially interchangeable, “essentially” being a word that admits a degree of latitude. In 2021, researchers J. Shorish, M.

Stephenson, and M. Zargham devised a mathematical model of fungibility (interchangeability) that would account for such variation. Distinguishing “exactly the same” and “nearly the same” commodities—whether crops, lumber, or precious metals—_____ among their primary aims.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. having been
- B. being
- C. was
- D. were

Question 21 Skill: Grammar / Conventions | Difficulty: Hard

That Chaucer’s *Canterbury Tales* inspired imitators is evident from the existence of three near-contemporary “continuations” of the collection: *The Siege of Thebes*, which purports to be a new tale told during the pilgrims’ _____ *The Tale of Beryn*, which depicts the pilgrims as tourists; and *The Ploughman’s Tale*, which features a minor character from the original work.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. return
- B. return,
- C. return.
- D. return;

Question 22 Skill: Transitions | Difficulty: Hard

Dorothea Lange and Walker Evans have both been recognized for their contributions to documentary photography during the Great Depression, though for different subjects. Lange was celebrated for her intimate portraits of displaced farmworkers; Evans, _____ was acclaimed for his architectural studies of rural Southern structures.

Which choice completes the text with the most logical transition?

- A. as a result,
- B. by contrast,
- C. in other words,
- D. for instance,

Question 23 Skill: Transitions | Difficulty: Hard

In a 2022 analysis of 200 terms, researchers found a broad pattern of usage-dependent change: words appearing less frequently in written texts saw a faster rate of connotative shift—_____ the rate at which a word’s implied associations will evolve over time. Adjectives (e.g., “noble”) saw the largest effect; prepositions (e.g., “within”), meanwhile, saw the smallest.

Which choice completes the text with the most logical transition?

- A. that is,
- B. for example,
- C. likewise,
- D. in addition,

Question 24 Skill: Transitions | Difficulty: Hard

At previous Venice Biennales, in contrast to pavilions showcasing internationally renowned contemporary artists, regional craft traditions were displayed as temporary side exhibitions. _____ while the internationally celebrated medium of glass sculpture became a permanent feature of the Murano district's cultural offerings after its initial Biennale demonstration, traditional Venetian mask-making, an art form popular mainly within the city, appeared only at the 1980 exhibition.

Which choice completes the text with the most logical transition?

- A. Moreover,
- B. Granted,
- C. Accordingly,
- D. However,

Question 25 *Skill: Rhetorical Synthesis | Difficulty: Hard*

While researching a topic, a student has taken the following notes:

- Japanese architect Tadao Ando designed the Church of the Light in Ibaraki, completed in 1989.
- The Church of the Light uses a cruciform aperture in one wall to admit natural sunlight, creating a cross of light on the interior wall.
- Ando also designed the Water Temple in Awaji, completed in 1991.
- The Water Temple uses a lotus pond as its entrance, requiring visitors to descend through water to reach the worship hall below.
- Both buildings are examples of Ando's minimalist concrete style.

The student wants to highlight a difference between the two buildings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Both the Church of the Light and the Water Temple exemplify Ando's minimalist concrete style, completed in 1989 and 1991 respectively.
- B. The Church of the Light uses a cruciform aperture to create a cross of light on its interior wall, whereas the Water Temple uses a lotus pond as its entrance, requiring visitors to descend through water.
- C. Ando designed both the Church of the Light in Ibaraki and the Water Temple in Awaji using minimalist concrete.
- D. The Water Temple, completed in 1991, was designed after the Church of the Light, completed in 1989, and both showcase Ando's style.

Question 26 *Skill: Rhetorical Synthesis | Difficulty: Hard*

While researching a topic, a student has taken the following notes:

- A high-eruptive-frequency volcano is one that erupts many times over a geologically short period, producing calderas and pyroclastic deposits.
- Mount Etna in Sicily has erupted over 200 times in recorded history.
- Mount Etna's frequent eruptions have created multiple summit calderas.
- Mauna Loa in Hawaii has erupted 33 times since 1843.
- Mauna Loa's eruptions are typically effusive, producing lava flows rather than explosive pyroclastic material.

The student wants to define the term “high-eruptive-frequency volcano.” Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. A high-eruptive-frequency volcano is one that erupts many times over a geologically short period, often producing calderas and pyroclastic deposits.
- B. Mount Etna, with over 200 recorded eruptions, is a high-eruptive-frequency volcano, unlike Mauna Loa, which has erupted only 33 times since 1843.
- C. A high-eruptive-frequency volcano erupts many times and is located in areas such as Sicily and Hawaii.
- D. Mount Etna’s eruptions have created multiple summit calderas, making it one of the most studied volcanoes in the world.

Question 27 *Skill: Rhetorical Synthesis | Difficulty: Hard*

While researching a topic, a student has taken the following notes:

- Saturn has at least 146 known moons, including Titan and Enceladus.
- Titan orbits Saturn in a prograde direction (the same direction as Saturn’s rotation) with a period of 15.95 Earth days.
- Enceladus orbits Saturn in a prograde direction with a period of 1.37 Earth days.
- Both moons’ orbits are nearly circular rather than highly elliptical.
- Enceladus is known for its subsurface ocean and ice geysers.

The student wants to contrast the orbits of the two moons. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A. Titan and Enceladus both orbit Saturn in a prograde direction, and both have nearly circular orbits.
 - B. Titan orbits Saturn with a period of 15.95 Earth days, whereas Enceladus is known for its subsurface ocean and ice geysers.
 - C. Saturn has at least 146 known moons, and both Titan and Enceladus orbit in a prograde direction.
 - D. Although both Titan and Enceladus orbit Saturn in a prograde direction with nearly circular orbits, Titan’s orbital period of 15.95 Earth days is substantially longer than Enceladus’s period of 1.37 Earth days.
-

Section 2: Math

Module 1 (22 Questions) - All Students

Time: 35 minutes

Question 1 *Skill: Exponential vs. Linear Growth | Difficulty: Medium*

The table shows selected values from the function g .

x	$g(x)$
-1	$1/52$
0	1
1	52
2	2,704

Which of the following is the best description of function g ?

A. Increasing linear

- B. Decreasing linear
- C. Increasing exponential
- D. Decreasing exponential

Question 2 Skill: Graphs of Linear Equations | Difficulty: Medium

The function g is defined by $g(x) = -5x + 30$. What is the value of $g(x)$ when $x = -4$?

- A. 50
- B. 45
- C. 10
- D. 30

Question 3 Skill: Linear Equations in One & Two Variables | Difficulty: Medium

$x^2 = (25)(25)$. What is the positive solution to the given equation?

Question 4 Skill: Statistics: Mean & Median | Difficulty: Medium

Data set A: 3, 3, 3, 6, 6, 9, 12, 12, 15, 15, 15 Data set B: 3, 6, 6, 9, 9, 9, 12, 12, 15, 15

Data set A and data set B each have 11 values. Which of the following statements best compares the median of data set A and the median of data set B?

- A. The median of data set A is greater than the median of data set B.
- B. The median of data set A is less than the median of data set B.
- C. The medians of data sets A and B are equal.
- D. There is not enough information to compare the medians.

Question 5 Skill: Polynomial Operations | Difficulty: Medium

The function f is defined by $f(x) = x^3 + 8$. What is the value of $f(3)$?

Question 6 Skill: Exponential Functions - Growth & Decay | Difficulty: Medium

In the xy -plane, the graph of the equation $y = 7^x$ is translated 5 units down. What is an equation of the resulting graph?

- A. $y = 7^x + 5$
- B. $y = 7^x - 5$
- C. $y = 7^{(x+5)}$
- D. $y = 7^{(x-5)}$

Question 7 Skill: Systems of Linear Equations | Difficulty: Medium

$$y = 4x + 12 \quad y = (x + 2)^2$$

A solution to the given system of equations is (x, y) . What is the value of x^2 ?

Question 8 Skill: Interpreting Coefficients in Linear Models | Difficulty: Medium

$$f(x) = -5x^2 + 70x$$

The function f models the height above ground level, in feet, of a certain firework x seconds after it is launched. What is the best interpretation of $f(2) = 120$ in this context?

- A. 120 seconds after the firework is launched, its estimated height is 2 feet above ground level.
- B. The firework reaches an estimated maximum height of 120 feet above ground level.
- C. 2 seconds after the firework is launched, its estimated height is 120 feet above ground level.
- D. Approximately 2 seconds after launching, the firework returns to ground level.

Question 9 Skill: Linear Equations in One & Two Variables | Difficulty: Medium

A school drama club sold tickets to raise money. The club raised \$8 from each student ticket sold and raised \$12 from each adult ticket sold. The club raised a total of \$296 from the sale of these tickets. The equation $8x + 12y = 296$ represents this situation. Which of the following is the best interpretation of y in this context?

- A. The number of adult tickets sold
- B. The number of student tickets sold
- C. The price, in dollars, of each adult ticket
- D. The price, in dollars, of each student ticket

Question 10 Skill: Systems of Linear Equations | Difficulty: Medium

A pharmacist mixed x milliliters of a 4% saline solution with y milliliters of a 10% saline solution to produce a 6% saline solution. Which equation best represents this situation? (Assume the volumes of the solutions are additive.)

- A. $0.04x + 0.10y = 6(x + y)$
- B. $0.04x + 0.10y = 0.06(x + y)$
- C. $0.4x + 1.0y = 6(x + y)$
- D. $0.4x + 1.0y = 0.6(x + y)$

Question 11 Skill: Right Triangle Trigonometry | Difficulty: Medium

[Figure: A right triangle with legs of length 24 and 32, hypotenuse c , and angle x° opposite the side of length 24. Note: Figure not drawn to scale.]

In the right triangle shown, c is a constant. The value of $\sin x^\circ = k/c$, where k is a constant. What is the value of k ?

Question 12 Skill: Data Interpretation from Tables and Graphs | Difficulty: Medium

GRAPH TO INSERT

[Graph: Bar graph showing the frequency of each data value in a data set. X-axis: Data value (1 through 6). Y-axis: Frequency (0 through 9). Bar heights: value 1 = 3, value 2 = 6, value 3 = 4, value 4 = 8, value 5 = 5, value 6 = 7.]

The bar graph shows the frequency of each data value in a data set. What is the frequency of data value 2 in this data set?

Question 13 Skill: Similar Triangles | Difficulty: Medium

The table gives the perimeters of similar triangles ABC and DEF, where AB corresponds to DE. The length of AB is 12.

	Perimeter
Triangle ABC	36
Triangle DEF	252

What is the length of DE?

- A. 3
- B. 12
- C. 48
- D. 84

Question 14 *Skill: Circle Equations | Difficulty: Medium*

What is the radius of the circle in the xy -plane defined by $(x + 3)^2 + (y + 5)^2 = 121$?

Question 15 *Skill: Creating Linear Models from Word Problems | Difficulty: Medium*

A piece of rope with a length of 84 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 10 more than 3 times the value of y . What is the value of x ?

- A. 18.5
- B. 22
- C. 65.5
- D. 74

Question 16 *Skill: Area & Perimeter | Difficulty: Medium*

The area of a rectangle is 180 square centimeters. The length of the rectangle is 8 centimeters greater than the width of the rectangle. What is the width, in centimeters, of the rectangle?

- A. 20
- B. 12
- C. 18
- D. 10

Question 17 *Skill: Linear Inequalities | Difficulty: Medium*

A consultant charges \$40 per hour for the first 30 hours worked in a week. For each hour worked over 30 hours, the consultant charges \$60 per hour. What is the least number of whole hours that the consultant must work in a week to earn at least \$1,900?

- A. 42
- B. 41
- C. 12
- D. 11

Question 18 *Skill: Nonlinear Systems of Equations | Difficulty: Medium*

[Graph: A coordinate plane showing the graph of a system of a linear equation (horizontal line at $y = 5$) and a quadratic equation (downward-opening parabola with vertex near $(5, 8)$). The two graphs

intersect at approximately $x = 2$ and $x = 8$. Y-axis from -1 to 10 , X-axis from 0 to 14 .]

The graph of a system of a linear equation and a quadratic equation is shown. A solution to the system is (x, y) . What is a possible value of x ?

- A. 2
- B. 5
- C. 7.5
- D. 9

Question 19 Skill: Systems of Equations - Number of Solutions | Difficulty: Medium

How many solutions does the equation $13(x - 12) = -15(x + 13)$ have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Question 20 Skill: Systems of Linear Equations | Difficulty: Medium

$$13x - 5y = 26 \quad 4x + 3y = 37$$

The solution to the given system of equations is $(x, y) = (p/59, w/59)$, where p and w are integers. What is the value of p ?

Question 21 Skill: Creating Linear Models from Word Problems | Difficulty: Medium

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y .

x	y
1	$-197/10$
3	$-179/10$
6	$-152/10$

What is the x -coordinate of the x -intercept of the graph that represents this linear relationship in the xy -plane?

Question 22 Skill: Similar Triangles | Difficulty: Medium

[Figure: Line segment LQ intersects line segment MP at point R . LM is parallel to PQ . The lengths of MR , LR , and RP are 6 , 8 , and 15 , respectively.]

In the figure, LQ intersects MP at point R , and LM is parallel to PQ . The lengths of MR , LR , and RP are 6 , 8 , and 15 , respectively. What is the length of LQ ?

- A. 20
 - B. 22
 - C. 26
 - D. 28
-

Section 2: Math

Module 2 - Easier Path (22 Questions)

Time: 35 minutes

Question 1 Skill: Linear Equations in One & Two Variables | Difficulty: Easy

If $36 = 8 + 4(x - 2)$, what is the value of $4(x - 2)$?

- A. 8
- B. 20
- C. 28
- D. 36

Question 2 Skill: Polynomial Operations | Difficulty: Easy

Which expression is equivalent to $(x^2 + 9)^2 + (x - 7)(x + 7)$?

- A. $x^4 + x^2 + 130$
- B. $x^4 + 12x^2 + 81$
- C. $x^4 + 19x^2 - 49$
- D. $x^4 + 19x^2 + 32$

Question 3 Skill: Graphs of Linear Equations | Difficulty: Easy

The graph of the polynomial function f in the xy -plane, where $y = f(x)$, passes through the point $(7, 11)$. What is the value of $f(7)$?

- A. 7
- B. 11
- C. 18
- D. 77

Question 4 Skill: Exponential Functions - Growth & Decay | Difficulty: Easy

What is the y -intercept of the graph of $y = 8(3)^{x+2}$ in the xy -plane?

- A. $(0, 8)$
- B. $(0, 24)$
- C. $(0, 72)$
- D. $(0, 5,832)$

Question 5 Skill: Linear Inequalities | Difficulty: Easy

$$y > 3x - 4$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

- A. $x: 2, 4, 7 \rightarrow y: 1, 7, 16$
- B. $x: 2, 4, 7 \rightarrow y: 3, 9, 18$
- C. $x: 2, 4, 7 \rightarrow y: 2, 8, 17$
- D. $x: 2, 4, 7 \rightarrow y: 5, 11, 20$

Question 6 Skill: Creating Linear Models from Word Problems | Difficulty: Easy

Number of wagons	Maximum number of passengers and crew
4	128
6	194
12	392

The table shows the linear relationship between the number of wagons, w , on a heritage railway train and the maximum number of passengers and crew, p , that the train can carry. Which equation represents the linear relationship between w and p ?

- A. $33w - p = -4$
- B. $33w - p = 4$
- C. $33p - w = -4$
- D. $33p - w = 4$

Question 7 Skill: Function Notation & Evaluation | Difficulty: Easy

The function h is defined by $h(x) = 3(7x - 8)$. What is the value of $h(8)$?

- A. 8
- B. 16
- C. 132
- D. 144

Question 8 Skill: Coordinate Geometry | Difficulty: Easy

Line q in the xy -plane is defined by $y = x + 1/3$ and passes through the points $(a, 0)$ and $(0, b)$, where a and b are constants. What is the value of a ?

Question 9 Skill: Quadratic Equations - Solving | Difficulty: Easy

$$x^2 + 4x = 60$$

What is one of the solutions to the given equation?

- A. $2 + \sqrt{64}$
- B. $-2 + \sqrt{64}$
- C. $-2 + \sqrt{56}$
- D. $2 + \sqrt{56}$

Question 10 Skill: Standard Deviation | Difficulty: Easy

Data set P	10	10	11	11	12	13	13	14	14	14
Data set Q	1	1	2	2	3	4	4	5	5	5

The values in data sets P and Q are shown in the table. The standard deviation of data set P is r , and the standard deviation of data set Q is t . Which of the following statements about the standard deviations of the data sets is true?

- A. $r < t$
- B. $r > t$
- C. $r = t$
- D. The relationship between r and t cannot be determined.

Question 11 Skill: Ratios, Rates & Proportions | Difficulty: Easy

The ratio 7 to x is equivalent to the ratio y to 6. Which equation represents x in terms of y ?

- A. $x = 7y/6$
- B. $x = 6y/7$
- C. $x = y/42$
- D. $x = 42/y$

Question 12 Skill: Exponential Functions - Growth & Decay | Difficulty: Easy

For an online quiz game, 360 points are awarded for a correct answer if a question is answered in less than 4 seconds from the time the question is asked. Each 4 seconds after the question is asked, the number of points awarded for a correct answer decreases by 25% of the number of points awarded for a correct answer in the previous 4 seconds. Which function gives the number of points awarded for a correct answer x seconds from the time the question is asked, where x is a multiple of 4?

- A. $f(x) = 360(0.75)^{x/4}$
- B. $f(x) = 360(0.75)^4x$
- C. $f(x) = 360(0.25)^{x/4}$
- D. $f(x) = 360(0.25)^4x$

Question 13 Skill: Creating Linear Models from Word Problems | Difficulty: Easy

A factory produces 7-inch, 5-inch, and 2-inch bolts. During a certain shift, the number of 7-inch bolts that the factory produces is 3 times the number n of 5-inch bolts, and the number of 2-inch bolts is 20. During this shift, the factory produces 80 bolts total. Which equation represents this situation?

- A. $7(3n) + 5n + 2(20) = 80$
- B. $7n + 5n + 2n = 80$
- C. $4n + 20 = 80$
- D. $3n + 20 = 80$

Question 14 Skill: Circle Equations | Difficulty: Easy

A circle in the xy -plane has its center at $(3, 4)$. Line m is tangent to this circle at the point $(a, -2)$, where a is a constant. The slope of line m is $5/6$. What is the value of a ?

- A. -2
- B. 8
- C. $53/3$
- D. $61/4$

Question 15 Skill: Coordinate Geometry | Difficulty: Easy

[Graph: A line segment in the xy -plane from approximately $(-6, 2)$ to $(6, 8)$, representing one leg of a right triangle. The area of this triangle is $36\sqrt{5}$ square units.]

The line segment shown in the xy -plane represents one of the legs of a right triangle. The area of this triangle is $36\sqrt{5}$ square units. What is the length, in units, of the other leg of this triangle?

- A. 12
- B. 24
- C. $3\sqrt{10}$
- D. $6\sqrt{10}$

Question 16 Skill: Triangle Properties | Difficulty: Easy

In triangle PQR and triangle STU, PQ and ST are each equal to 8 centimeters, and angles P and S each have measure 40° . Which additional piece of information is sufficient to prove that triangle PQR is congruent to triangle STU?

- A. The lengths of sides QR and TU are equal.

- B. The lengths of sides PR and SU are equal.
- C. The perimeters of triangles PQR and STU are equal.
- D. No additional information is necessary to prove that the two triangles are congruent.

Question 17 Skill: Interpreting Coefficients in Linear Models | Difficulty: Easy

[Graph: A coordinate plane showing a line with positive slope, passing through approximately $(-6, -10)$ and $(4, 10)$. The y-intercept appears to be positive and the x-intercept appears to be negative.]

The graph of the linear function $y = f(x) - 15$ is shown. If c and d are positive constants, which equation could define f ?

- A. $f(x) = d - cx$
- B. $f(x) = -d + cx$
- C. $f(x) = d + cx$
- D. $f(x) = -d - cx$

Question 18 Skill: Probability - Basic | Difficulty: Easy

For 80 insects, the table summarizes the distribution of order classification and wing length.

Classification	Less than 15 mm	15 to 25 mm	Greater than 25 mm
Coleoptera	11	6	4
Lepidoptera	0	12	16
Hymenoptera	8	20	3

One of these insects will be selected at random. What is the probability of selecting an insect with a wing length that is less than or equal to 25 mm, given that it is not classified as Lepidoptera? (Express your answer as a decimal or fraction, not as a percent.)

Question 19 Skill: Radical and Rational Exponents | Difficulty: Easy

$$\sqrt{k - x} = 28 - x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

Question 20 Skill: Factoring Polynomials | Difficulty: Easy

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

- A. $3x^2 + 7x + 14b$
- B. $3x^2 + 16x + 14b$
- C. $3x^2 + 20x + 14b$
- D. $3x^2 + 25x + 14b$

Question 21 Skill: Discriminant & Number of Solutions | Difficulty: Easy

$$x(kx - 36) = -10$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k ?

Question 22 *Skill: Ratios, Rates & Proportions | Difficulty: Easy*

The mass of object X is 320% of the mass of object Y, and the mass of object X is 0.080% of the mass of object Z. If the mass of object Z is $p\%$ of the mass of object Y, what is the value of $p/1,000$?

Section 2: Math

Module 2 - Harder Path (22 Questions)

Time: 35 minutes

Question 1 Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

In the xy -plane, the graph of the equation $y = 4^x$ is translated 3 units to the left and then 7 units up. What is an equation of the resulting graph?

- A. $y = 4^{(x-3)} + 7$
- B. $y = 4^{(x+3)} - 7$
- C. $y = 4^{(x+3)} + 7$
- D. $y = 4^x + 10$

Question 2 Skill: Polynomial Operations | Difficulty: Hard

Which expression is equivalent to $(x^2 + 7)^2 + (x - 5)(x + 5)$?

- A. $x^4 + 14x^2 + 24$
- B. $x^4 + 14x^2 + 49$
- C. $x^4 + 15x^2 - 25$
- D. $x^4 + 15x^2 + 24$

Question 3 Skill: Systems of Linear Equations | Difficulty: Hard

$$y = 6x + 18 \quad y = (x + 3)^2$$

A solution to the given system of equations is (x, y) . What is the value of x^2 ?

- A. 3
- B. 9
- C. 12
- D. 18

Question 4 Skill: Quadratic Functions - Graphs & Properties | Difficulty: Hard

GRAPH TO INSERT

The function g is defined by $g(x) = -3x^2 + 54x - 192$. What is the value of x when $g(x)$ reaches its maximum?

- A. 9
- B. 12
- C. 51
- D. 54

Question 5 Skill: Linear Inequalities | Difficulty: Hard

$$y > 4x - 7$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

- A. $x: 2, 5, 9 \rightarrow y: 1, 12, 28$
- B. $x: 2, 5, 9 \rightarrow y: 4, 16, 30$
- C. $x: 2, 5, 9 \rightarrow y: 2, 14, 29$
- D. $x: 2, 5, 9 \rightarrow y: 3, 15, 29$

Question 6 Skill: Creating Linear Models from Word Problems | Difficulty: Hard

Number of carriages	Maximum number of passengers
3	123
5	201
8	318

The table shows the linear relationship between the number of carriages, c , on a commuter train and the maximum number of passengers, p , that the train can carry. Which equation represents the linear relationship between c and p ?

- A. $39c - p = 6$
- B. $39c + p = 6$
- C. $p - 39c = 6$
- D. $p + 39c = 6$

Question 7 Skill: Function Notation & Evaluation | Difficulty: Hard

The function f is defined by $f(x) = 2x^2 - 3x + 1$. What is the value of $f(-2) - f(1)$?

- A. 14
- B. 12
- C. 10
- D. 15

Question 8 Skill: Coordinate Geometry | Difficulty: Hard

Line p in the xy -plane is defined by $y = x + \frac{5}{8}$ and passes through the points $(a, 0)$ and $(0, b)$, where a and b are constants. What is the value of a ?

Question 9 Skill: Quadratic Equations - Solving | Difficulty: Hard

$$x^2 + 10x = 96$$

What is one of the solutions to the given equation?

- A. $5 + \sqrt{121}$
- B. $-5 + \sqrt{121}$
- C. $-5 + \sqrt{71}$
- D. $5 + \sqrt{71}$

Question 10 Skill: Standard Deviation | Difficulty: Hard

Data set	R	14	14	15	15	16	17	17	18	18	18
Data set S		5	5	6	6	7	8	8	9	9	9

The values in data sets R and S are shown. The standard deviation of data set R is u , and the standard deviation of data set S is v . Which of the following statements is true?

- A. $u < v$
- B. $u > v$
- C. $u = v$
- D. The relationship between u and v cannot be determined.

Question 11 Skill: Ratios, Rates & Proportions | Difficulty: Hard

The ratio 11 to x is equivalent to the ratio y to 8. Which equation represents x in terms of y ?

- A. $x = 11y/8$
- B. $x = 8y/11$
- C. $x = y/88$
- D. $x = 88/y$

Question 12 Skill: Exponential Functions - Growth & Decay | Difficulty: Hard

For an online quiz game, 800 points are awarded for a correct answer if a question is answered in less than 3 seconds from the time the question is asked. Each 3 seconds after the question is asked, the number of points awarded for a correct answer decreases by 30% of the number of points awarded for a correct answer in the previous 3 seconds. Which function gives the number of points awarded for a correct answer x seconds from the time the question is asked, where x is a multiple of 3?

- A. $f(x) = 800(0.70)^{x/3}$
- B. $f(x) = 800(0.70)^3x$
- C. $f(x) = 800(0.30)^{x/3}$
- D. $f(x) = 800(0.30)^3x$

Question 13 Skill: Creating Linear Models from Word Problems | Difficulty: Hard

A warehouse stores large, medium, and small crates. During a certain delivery, the number of large crates received is 5 times the number n of medium crates, and the number of small crates is 30. During this delivery, the warehouse receives 120 crates total. Which equation represents this situation?

- A. $5(5n) + 3n + 30 = 120$
- B. $5n + n + 30 = 120$
- C. $6n + 30 = 120$
- D. $5n + 30 = 120$

Question 14 Skill: Circle Equations | Difficulty: Hard

A circle in the xy -plane has its center at $(7, 2)$. Line m is tangent to this circle at the point $(a, -5)$, where a is a constant. The slope of line m is $4/3$. What is the value of a ?

- A. $7/4$
- B. $28/3$
- C. $49/3$
- D. $61/4$

Question 15 Skill: Coordinate Geometry | Difficulty: Hard

[Graph: A line segment in the xy-plane from (-3, 1) to (5, 5), representing one leg of a right triangle. The area of this triangle is $24\sqrt{5}$ square units.]

The line segment shown in the xy-plane represents one of the legs of a right triangle. The area of this triangle is $24\sqrt{5}$ square units. What is the length, in units, of the other leg of this triangle?

Question 16 Skill: Triangle Properties | Difficulty: Hard

In triangle MNO and triangle PQR, MN and PQ are each equal to 10 centimeters, and angles M and P each have measure 55° . Which additional piece of information is sufficient to prove that triangle MNO is congruent to triangle PQR?

- A. The lengths of sides MO and PR are equal.
- B. The perimeters of triangles MNO and PQR are equal.
- C. The lengths of sides NO and QR are equal.
- D. No additional information is necessary to prove that the two triangles are congruent.

Question 17 Skill: Probability - Basic | Difficulty: Hard

For 85 minerals, the table summarizes the distribution of hardness classification and crystal system.

Classification	Monoclinic	Orthorhombic	Hexagonal
Silicates	14	9	6
Oxides	0	10	18
Carbonates	7	18	3

One of these minerals will be selected at random. What is the probability of selecting a mineral with a Monoclinic or Orthorhombic crystal system, given that it is not classified as an Oxide?

- A. $37/57$
- B. $37/85$
- C. $48/85$
- D. $48/57$

Question 18 Skill: Radical and Rational Exponents | Difficulty: Hard

$$\sqrt{k - x} = 36 - x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

Question 19 Skill: Factoring Polynomials | Difficulty: Hard

Which of the following expressions has a factor of $x + 5b$, where b is a positive integer constant?

- A. $4x^2 + 16x + 20b$
- B. $4x^2 + 24x + 20b$
- C. $4x^2 + 32x + 20b$
- D. $4x^2 + 40x + 20b$

Question 20 Skill: Discriminant & Number of Solutions | Difficulty: Hard

$$x(kx - 54) = -18$$

In the given equation, k is an integer constant. If the equation has two distinct real solutions, what is the greatest possible value of k ?

Question 21 *Skill: Systems of Linear Equations | Difficulty: Hard*

$$17x - 4y = 31 \quad 3x + 11y = 52$$

The solution to the given system of equations is $(x, y) = (p/199, w/199)$, where p and w are integers. What is the value of p ?

Question 22 *Skill: Ratios, Rates & Proportions | Difficulty: Hard*

The mass of object P is 450% of the mass of object Q, and the mass of object P is 0.060% of the mass of object R. If the mass of object R is $s\%$ of the mass of object Q, what is the value of $s/1,000$?
